

Notes

In the following results, the Levenshtein distance tolerance of 0, 1 and 3 has been applied to the fields: Dates, address, buyer. The others are too short for this to be useful.

OCR + LLN

CHANGING THE TOLERANCE FOR THE EDIT DISTANCE SEEMS TO NOT INFLUENCE THE MISTAKES

Common errors

- Says 0.0 when a field is not in the layout even though in the plot it is explicitly said that 'if field does not appear, return "None" '.

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['discount', 'tax'].
Field | Type | Value

buyer | Gt | Raymond Baker
      | Pred | Raymond Baker
address | Gt | 924 Lin Trafficway Suite 687 Jackmouth, CA 90936 US
      | Pred | 924 Lin Trafficway Suite 687
      | Pred | Jackmouth, CA 90936 US
date | Gt | 10-Oct-2004
     | Pred | 2004-10-10
currency | Gt | USD
        | Pred | USD
subtotal | Gt | 390.5
         | Pred | 390.5
discount | Gt | None
         | Pred | 0.0
tax | Gt | None
   | Pred | 0.0
total | Gt | 400.43
     | Pred | 400.43

Current file: Template8_Instance195.jpg
```

- Duplicated labels when one is missing:

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['subtotal', 'discount'].
Field | Type | Value

buyer | Gt | Tara Barnes
      | Pred | Tara Barnes
address | Gt | 3732 Wright Ports Ashleyport, VT 08388 US
      | Pred | 3732 Wright Ports, Ashleyport, VT 08388 US
date | Gt | 01-Jun-2012
     | Pred | 2012-06-01
currency | Gt | USD
        | Pred | USD
subtotal | Gt | None
         | Pred | 902.62
discount | Gt | None
         | Pred | 0.0
tax | Gt | 56.14
   | Pred | 56.14
total | Gt | 902.62
     | Pred | 902.62

Current file: Template3_Instance121.jpg
```

- Gets the address of a different thing that is not the buyer's

```
... Current folder: orc_llm_keep/prebuilt-read
Failed at ['address'].
Field | Type | Value

buyer | Gt | Janice Nguyen
      | Pred | Janice Nguyen
address | Gt | 193 Troy Radial Crossborough, NV 81506 US
      | Pred | 16424 Timothy Mission, Markville, AK 58294 US
date | Gt | 11-Aug-2017
     | Pred | 2017-08-11
currency | Gt | $
        | Pred | $
subtotal | Gt | 286.76
         | Pred | 286.76
discount | Gt | 12.7
         | Pred | 12.7
tax | Gt | 15.34
   | Pred | 15.34
total | Gt | 291.69
     | Pred | 291.69

Current file: Template1_Instance11.jpg
```



INVOICE
Invoice Date: 11-Aug-2017
Due Date: 23-May-2000
PO Number: 99

Address: 16424 Timothy Mission
Markville, AK 58294 US

Email: melvin40@example.net
www.ThompsonandSons.org
(GSTIN: 12345670 00070007)

Buyer: Janice Nguyen
193 Troy Radial
Crossborough, NV 81506 US
Tel: + (908) 834-1601
Email: johnsonkarina@example.org
Site: http://livingston.com/

GSTIN: OG@AAMFC0376K124

Edit Distance 0

	Field	Hits	Proportion	Acuracy	Precision	Recall	F_score
General Score		41/100	0.4100	0.8862	0.8364	0.8395	0.8379
	buyer	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	address	99/100	0.9900	0.8794	0.8794	0.8794	0.8794
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	80/100	0.8000	0.8000	0.8000	0.8000	0.8000
	discount	44/100	0.4400	0.4400	0.4400	0.4400	0.4400
	tax	86/100	0.8600	0.8600	0.8600	0.8600	0.8600
	total	100/100	1.0000	1.0000	1.0000	1.0000	1.0000

Edit Distance 1

	Field	Hits	Proportion	Acuracy	Precision	Recall	F_score
General Score		41/100	0.4100	0.8862	0.8364	0.8395	0.8379
	buyer	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	address	99/100	0.9900	0.8794	0.8794	0.8794	0.8794
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	80/100	0.8000	0.8000	0.8000	0.8000	0.8000
	discount	44/100	0.4400	0.4400	0.4400	0.4400	0.4400
	tax	86/100	0.8600	0.8600	0.8600	0.8600	0.8600
	total	100/100	1.0000	1.0000	1.0000	1.0000	1.0000

Edit Distance 3

	Field	Hits	Proportion	Acuracy	Precision	Recall	F_score
General Score		41/100	0.4100	0.8862	0.8364	0.8395	0.8379
	buyer	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	address	99/100	0.9900	0.8794	0.8794	0.8794	0.8794
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	80/100	0.8000	0.8000	0.8000	0.8000	0.8000
	discount	44/100	0.4400	0.4400	0.4400	0.4400	0.4400
	tax	86/100	0.8600	0.8600	0.8600	0.8600	0.8600
	total	100/100	1.0000	1.0000	1.0000	1.0000	1.0000

OCR FINETUNED 5x5

Nearly all the scores are the same regardless of how many training files used in train

Common errors

- Adds 'to:' before the name of the buyer: 'to:Name' instead of 'Name'

```

Current folder: orc_llm_keep/ocr_finetuned_2x5_v1
Failed at ['buyer'].
Field | Type | Value

buyer | Gt | Janet West
      | Pred | to:Janet West
address | Gt | 490 Melinda Streets Apt. 806 Lauramouth, ID 47708 US
      | Pred | 490 Melinda Streets Apt. 806 Lauramouth, ID 47708 US
date | Gt | 31-May-1999
      | Pred | 1999-05-31
currency | Gt | USD
      | Pred | USD
subtotal | Gt | 735.75
      | Pred | 735.75
discount | Gt | 10.23
      | Pred | 10.23
tax | Gt | 42.69
      | Pred | 42.69
total | Gt | 749.16
      | Pred | 749.16

Current file: Template1_Instance24.jpg

```

- Adds the currency:'xxxx EUR' instead of just 'xxxx'

```

Current folder: orc_llm_keep/ocr_finetuned_2x5_v1
Failed at ['tax', 'total'].
Field | Type | Value

buyer | Gt | Cheryl Morgan
      | Pred | Cheryl Morgan
address | Gt | 1769 Mason Crossing Lindsayfort, AL 35116 US
      | Pred | 1769 Mason Crossing Lindsayfort, AL 35116 US
date | Gt | 14-Dec-2008
      | Pred | 2008-12-14
currency | Gt | EUR
      | Pred | EUR
subtotal | Gt | None
      | Pred | None
discount | Gt | None
      | Pred | None
tax | Gt | 27.55
      | Pred | 27.55 EUR
total | Gt | 426.54
      | Pred | 426.54 EUR

Current file: Template3_Instance185.jpg

```

IN THIS CASE BOTH ERROR COULD BE EASILY CORRECTED WITH LITTLE POST-PROCESSING

Edit Distance 0

	Field	Hits	Proportion	Acuracy	Precision	Recall	F_score
General Score		78/100	0.7800	0.9725	0.9111	0.9111	0.9111
	buyer	81/100	0.8100	0.9050	0.9050	0.9050	0.9050
	address	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	100/100	1.0000	0.9000	0.9000	0.9000	0.9000
	discount	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	tax	97/100	0.9700	0.9400	0.9250	0.9400	0.9300
	total	100/100	1.0000	0.9200	0.9200	0.9200	0.9200

Edit Distance 1

	Field	Hits	Proportion	Accuracy	Precision	Recall	F_score
General Score		79/100	0.7900	0.9738	0.9111	0.9111	0.9111
	buyer	82/100	0.8200	0.9050	0.9050	0.9050	0.9050
	address	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	100/100	1.0000	0.9000	0.9000	0.9000	0.9000
	discount	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	tax	97/100	0.9700	0.9400	0.9250	0.9400	0.9300
	total	100/100	1.0000	0.9200	0.9200	0.9200	0.9200

Edit Distance 3

	Field	Hits	Proportion	Accuracy	Precision	Recall	F_score
General Score		97/100	0.9700	0.9962	0.9111	0.9111	0.9111
	buyer	100/100	1.0000	0.9050	0.9050	0.9050	0.9050
	address	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	date	100/100	1.0000	0.0000	0.0000	0.0000	0.0000
	currency	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	subtotal	100/100	1.0000	0.9000	0.9000	0.9000	0.9000
	discount	100/100	1.0000	1.0000	1.0000	1.0000	1.0000
	tax	97/100	0.9700	0.9400	0.9250	0.9400	0.9300
	total	100/100	1.0000	0.9200	0.9200	0.9200	0.9200

DONUT FINETUNED 5x5

Common errors:

- Not finding the fields:

```
Current folder: donut_v2/donut_comp_5x5_30_100
Failed at ['date', 'currency', 'subtotal'].
Field | Type | Value
buyer | Gt | Shannon Alexander
address | Gt | 838 Harris Ferry Matthewburgh, MP 10645 US
date | Gt | 17-Sep-2009
currency | Gt | USD
subtotal | Gt | 707.98
discount | Gt | None
tax | Gt | None
total | Gt | 722.84
Current file: Template8_Instance140.jpg
```

- Duplicated labels when one is missing:

```
Failed at ['address'].
Failed at ['subtotal', 'total'].
Field | Type | Value

buyer | Gt | Gary Cross
      | Pred | Gary Cross
address | Gt | 5528 Vickie Fords South Heatherside, NH 42370 US
      | Pred | 5528 Vickie Fords South Heatherside, NH 42370 US
date | Gt | 26-Feb-2003
      | Pred | 26-Feb-2003
currency | Gt | EUR
      | Pred | EUR
subtotal | Gt | None
      | Pred | 658.69
discount | Gt | None
      | Pred | None
tax | Gt | 37.06
      | Pred | 37.06
total | Gt | 656.69
      | Pred | 658.69

Current file: Template3_Instance156.jpg
```

- Wrong numbers

```
Current folder: donut_v2/donut_comp_5x5_30_100
Failed at ['address', 'total'].
Field | Type | Value

buyer | Gt | Jason Costa
      | Pred | Jason Costa
address | Gt | 508 Schmidt Ramp Roseview, WI 79683 US
      | Pred | 500 Schmidt Ramp Roseview, WI 90688 US
date | Gt | 19-Jun-2012
      | Pred | 19-Jun-2012
currency | Gt | EUR
      | Pred | EUR
subtotal | Gt | 784.29
      | Pred | 784.29
discount | Gt | None
      | Pred | None
tax | Gt | 45.88
      | Pred | 45.88
total | Gt | 812.13
      | Pred | 812.00

Current file: Template5_Instance153.jpg
```

Edit Distance 0

	Field	Hits	Proportion	Accuracy	Precision	Recall	F_score
General Score		50/100	0.5000	0.9175	0.9182	0.8814	0.8930
	buyer	88/100	0.8800	0.9200	0.9200	0.9200	0.9200
	address	79/100	0.7900	0.8581	0.8666	0.8581	0.8615
	date	99/100	0.9900	0.9100	0.9100	0.9100	0.9100
	currency	99/100	0.9900	0.9900	0.9900	0.9900	0.9900
	subtotal	86/100	0.8600	0.7600	0.7600	0.7600	0.7600
	discount	97/100	0.9700	0.9700	0.9700	0.9700	0.9700
	tax	94/100	0.9400	0.8800	0.8800	0.8800	0.8800
	total	92/100	0.9200	0.8400	0.8400	0.8400	0.8400

Edit Distance 1

	Field	Hits	Proportion	Accuracy	Precision	Recall	F_score
General Score		51/100	0.5100	0.9175	0.9182	0.8814	0.8930
	buyer	95/100	0.9500	0.9200	0.9200	0.9200	0.9200
	address	72/100	0.7200	0.8581	0.8666	0.8581	0.8615
	date	99/100	0.9900	0.9100	0.9100	0.9100	0.9100
	currency	99/100	0.9900	0.9900	0.9900	0.9900	0.9900
	subtotal	86/100	0.8600	0.7600	0.7600	0.7600	0.7600
	discount	97/100	0.9700	0.9700	0.9700	0.9700	0.9700
	tax	94/100	0.9400	0.8800	0.8800	0.8800	0.8800
	total	92/100	0.9200	0.8400	0.8400	0.8400	0.8400

Edit Distance 3

		Field	Hits	Proportion	Acuracy	Precision	Recall	F_score
General Score			64/100	0.6400	0.9387	0.9182	0.8814	0.8930
		buyer	96/100	0.9600	0.9200	0.9200	0.9200	0.9200
		address	88/100	0.8800	0.8581	0.8666	0.8581	0.8615
		date	99/100	0.9900	0.9100	0.9100	0.9100	0.9100
		currency	99/100	0.9900	0.9900	0.9900	0.9900	0.9900
		subtotal	86/100	0.8600	0.7600	0.7600	0.7600	0.7600
		discount	97/100	0.9700	0.9700	0.9700	0.9700	0.9700
		tax	94/100	0.9400	0.8800	0.8800	0.8800	0.8800
		total	92/100	0.9200	0.8400	0.8400	0.8400	0.8400